POOJA VISWANATHAN

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PROFILE

CS undergraduate also pursuing a Minor in MedTech, with a strong interest in health technology, clinical software systems, and computational biology. My work spans building a clinical-grade sarcopenia detection model using surface EMG signals and custom CNN–Transformer architectures, and designing a full emergency-department clinical workflow system with medico-legal compliance and audit safeguards.

EDUCATION

PES University, Bangalore *2023 – 2027*

B.Tech in Computer Science & Engineering | CGPA: 8.11

Minor in MedTech (in progress) — Algorithms in Medicine, Healthcare Informatics, Radiology and Imaging, Healthcare Analytics

Completed a hospital internship at PES Hospital (Emergency & Casualty department), observing live clinical workflows and gaining hands-on insight into diagnostic equipment, patient triage processes, and the operational challenges physicians face in a real emergency-care setting.

Gear Innovative International School *2018 – 2023*

RESEARCH & PROJECTS

Emergency Department Triage & Medico-Legal Workflow System *2026*

- Built a production-style Emergency Department console covering structured AIIMS-TP 5-level triage, treat-first quick registration, and India's medico-legal (MLC) workflow, with session-based auth, role-based access control (RBAC), and a tamper-evident SHA-256 hash-chained audit log.
- Engineered the triage advisory logic to recompute server-side on submission so the advisory-only clinical guarantee cannot be bypassed by editing client code, storing both suggested and clinician-confirmed acuity levels for monthly override reporting.
- Implemented a mandatory-reporting engine (POCSO / IDSP / burns / poisoning) with statute citations, a full chain-of-custody evidence log, and all 6 disposition pathways with database-enforced mandatory fields per type.
- Designed the schema and workflows around healthcare compliance mandates; verified with 106 automated tests across 4 suites (end-to-end, RBAC, workflow, and functional coverage).

Live demo: ed-triage-snowy.vercel.app

Sarcopenia Detection Using sEMG (TCCT Model) *2025-2026*

- Designed a dual-branch deep learning architecture combining CNNs and Transformers with a custom cross-attention fusion mechanism (LCAF) for adaptive, bidirectional learning between raw EMG signals and domain-specific biomedical features.
- Achieved 92.2% classification accuracy and $R^2 = 0.97$ on subject-independent validation, demonstrating strong generalizability for clinical screening applications.
- Engineered the model for lightweight, wearable deployment, enabling real-time sarcopenia risk assessment from surface electromyography data, bridging computational methods with musculoskeletal health diagnostics.

Emotion-Aware Multi-Voice Audiobook Generation System *2025*

- Built an end-to-end generative AI pipeline converting text into expressive, multi-speaker audiobooks using LLM-based dialogue parsing and emotion classification.
- Implemented dynamic voice mapping and emotion-aware speech synthesis with edge-TTS and diffusion-based sound effect generation via HuggingFace AudioLDM, with a modular pipeline for audio stitching and narration output.

TECHNICAL SKILLS

Healthcare Software & Clinical Systems: Clinical Workflow Design, RBAC & Audit Logging, Healthcare Compliance (POCSO, BNSS, DPDP, NABH), Flask, PostgreSQL, Session-Based Auth

Medical Imaging & Health Informatics: DICOM & PACS, Imaging Modalities (X-ray, CT, MRI, Ultrasound), Image Processing (Sobel Filtering, Otsu Thresholding, Morphological Operations), Clinical Decision Algorithms, HL7/FHIR & EHR Data Standards, Healthcare Analytics & Biostatistics

Machine Learning & AI: Deep Learning (CNNs, Transformers, Cross-Attention Mechanisms), Scikit-learn, Signal Classification & Regression, Subject-Independent Validation, Model Optimization for Edge Deployment

Programming & Data Analysis: Python, C, Java, NumPy, Pandas, Matplotlib, Jupyter Notebooks, Data Structures & Algorithms

Tools & Platforms: Git, GitHub, Linux (Ubuntu), MERN Stack, Cisco Packet Tracer, Mininet

RESEARCH INTERESTS

Computational Biology, Bioinformatics, Healthcare Informatics, Clinical Decision Support Systems, Medical Imaging, Biological Network Analysis, AI in Healthcare, Genomics & Proteomics, Biomedical Signal Processing, Machine Learning for Drug Discovery

AWARDS & HONORS

- Prof. CNR Rao Merit Scholarship — Recipient (5th and 6th Semester), awarded for academic excellence at PES University.
- Mini Nobel Prize — Gear Innovative International School, for excellence in science and innovation.
- School Blue Award — Recognized for all-round performance in academics and extracurriculars.

LANGUAGES

English, Hindi, Tamil